

FINAL YEAR PROJECT REPORT

Fixing the Gaps

A Stakeholder-Driven Power BI Dashboard for Resource Allocation and School Performance in Pakistan

Business Intelligence & Advanced Data Analytics

Capstone Project Documentation

Prepared for: Government Education Decision-Making Context

Technology: Microsoft Power BI, Power Query, DAX

Dataset: Pakistan Education Performance Dataset

Source: <https://www.kaggle.com/code/jihyeseo/pakistan-school-data-eda/notebook>

Data Period: 2013–2016

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1. Introduction

1.1 Project Overview

This project develops an interactive Power BI reporting solution for analyzing school performance, infrastructure, enrollment, retention and policy context in Pakistan. It is designed to help government education stakeholders move from high-level monitoring to evidence-based identification and prioritization of intervention areas.

1.2 Problem Statement

Educational data can be difficult to convert into actionable decisions when performance, infrastructure, gender participation, geography and contextual indicators are viewed separately. This project addresses that pain point by combining these indicators into a structured, interactive decision-support environment.

1.3 Project Objectives

1. Consolidate education, infrastructure, geographic and contextual indicators into a structured Power BI model.
2. Provide an executive-level view of education performance.
3. Identify regional and district-level disparities.
4. Detect gender enrollment and infrastructure-related risks.
5. Analyze education, learning and retention trends.
6. Identify districts requiring priority intervention.
7. Provide policy and socio-economic context for interpreting education outcomes.
8. Use advanced Power BI visuals and interactivity to support faster decision making.
9. Translate analytical findings into recommended government actions.

2. Stakeholders and Intended Users

Persona	Need	Dashboard
Senior Education Executives	Rapid overview and major risks	Executive Governance Panel
Policy/Risk Decision Makers	Risk identification and priority intervention	Executive Risk & Intervention
District/Operational Officers	Infrastructure and district investigation	District Infrastructure Audit
Policy Analysts	Longitudinal and socio-economic context	Policy Impact & Socio-Economic
BI/Monitoring Teams	Detailed anomaly investigation	Analysis / Intervention Views

3. Data Source and Dataset

The project uses the Pakistan Education Performance Dataset sourced from Kaggle. The available data covers 2013–2016 and contains education performance, enrollment, infrastructure, geographic and contextual indicators.

- Boys and girls enrollment
- Complete primary schools
- Single-classroom and single-teacher schools
- All/partial/no facility indicators
- Electricity, drinking water and toilets
- Building condition
- Total schools
- Province and city information
- Year and national contextual indicators

4. Data Model and Power BI Architecture

The model separates the education fact table from time, geography and context dimensions, supporting reusable DAX measures, filtering, drill-down and drill-through.

Table	Purpose
Fact_Education_Metrics	Core education, enrollment, infrastructure and outcome measurements
Dim_Time	Year dimension for temporal analysis
Dim_Geography	Province, city, population, area and location information
Dim_Context	National policy and socio-economic context
Measures	Reusable DAX measures and risk calculations

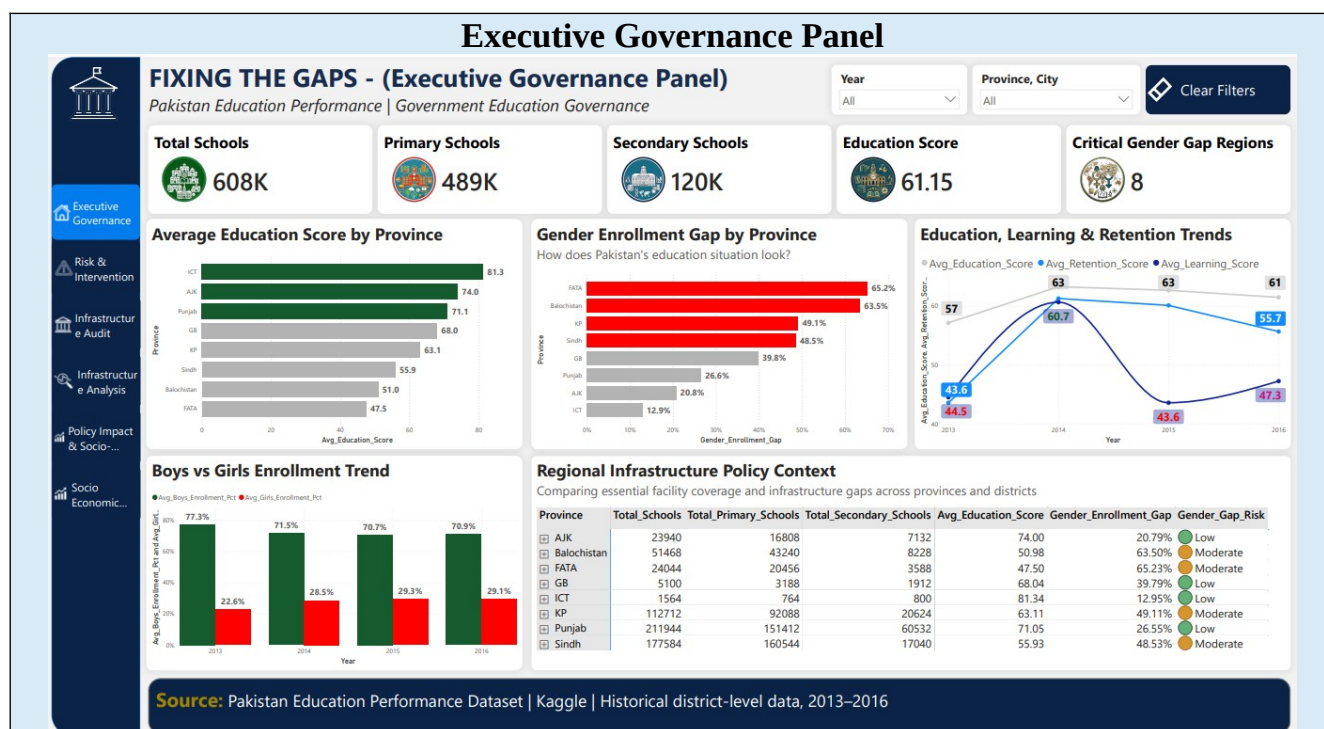
5. Key Measures and Indicators

Measure	Purpose
Avg_Boys_Enrollment_Pct	Average boys enrollment
Avg_Girls_Enrollment_Pct	Average girls enrollment
Gender_Enrollment_Gap	Difference between boys and girls enrollment
Utility_Deficit_Rate	Share represented by No Facility
Single_Teacher_Strain_Rate	Teacher-capacity pressure
Single_Classroom_Strain_Rate	Classroom-capacity pressure
Avg_Education_Score	Education performance

Avg_Learning_Score	Learning performance
Avg_Retention_Score	Retention performance
Avg_Electricity / Avg_Drinking_Water / Avg_Toilet	Facility coverage indicators
Overall_Risk_Score	Maximum of relevant risk scores

6. Dashboard 1 — Executive Governance Panel

The Executive Governance Panel is the main executive-facing dashboard. It provides a compact view of the overall education situation and highlights areas requiring deeper investigation.



6.1 Questions Answered

- How many schools are represented?
- What is the overall education performance?
- Which regions show critical gender gaps?
- Which regions show critical infrastructure conditions?
- How do education, learning and retention change over time?
- How large is the boys-versus-girls enrollment gap?

6.2 Main Visuals

- KPI cards
- Critical Gender Gap Regions card
- Critical Infrastructure Regions card
- Education Score by Province

- Gender Enrollment Gap by Province
- Education, Learning & Retention Trends
- Boys vs Girls Enrollment Trend

6.3 Executive Risk & Intervention



This supporting page adds anomaly detection, risk scoring, a Priority Intervention Matrix, a gauge for Overall Risk Score, priority region indicators and recommended actions.

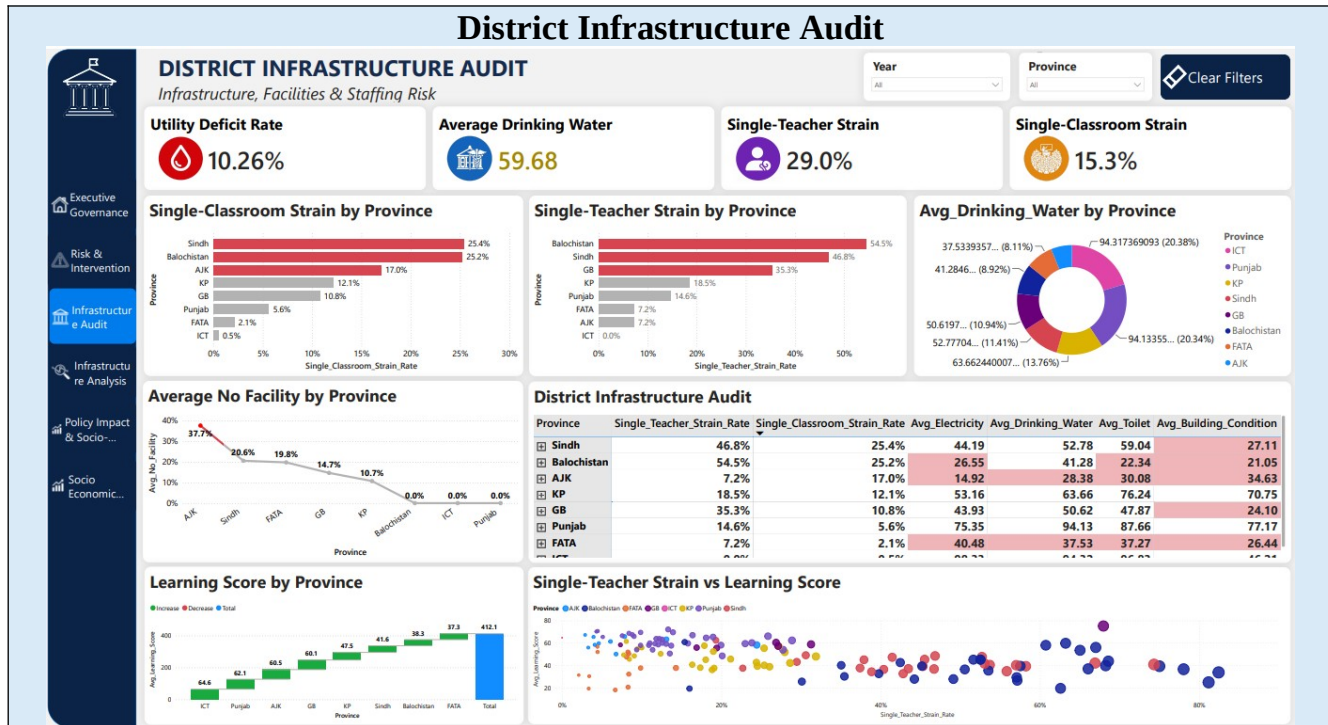
Kohistan Indicator	Value
Gender Enrollment Gap	93.0%
Gender Gap Risk	Critical
Utility Deficit	37.89%
Single Teacher Strain	28.6%
Single Classroom Strain	20.5%
Infrastructure Risk	High
Learning Score	42.1
Retention Score	30.0

Recommended Government Action

Prioritize an integrated intervention in Kohistan combining gender participation measures, essential facility improvement, teacher and classroom capacity support, learning support and retention measures. Monitor annual indicators to evaluate progress.

7. Dashboard 2 — District Infrastructure Audit

The District Infrastructure Audit focuses on school facility coverage and district-level infrastructure deficiencies for operational and regional users.



7.1 Questions Answered

- Which provinces and cities have the largest infrastructure deficits?
- Where are electricity, drinking water and toilet facilities weakest?
- Which districts have high teacher strain?
- Which districts have high classroom strain?
- Which regions have High or Critical infrastructure risk?
- Which districts combine infrastructure deficiencies with weak outcomes?

7.2 Advanced Visuals and Interactivity

- Decomposition Tree
- Treemap
- Province → City hierarchy
- Drill-through
- X/Y average reference lines
- Conditional formatting

- Infrastructure risk matrix

7.3 Key Findings

District	No Facility	Education	Learning	Retention
Kohistan	37.89%	40.1	42.1	30.0
Thatta	44.42%	45.6	36.7	24.3
Tharparkar	52.89%	51.1	34.8	33.1
Haveli	56.69%	72.7	63.2	55.6
Hattian	53.74%	71.2	65.3	55.6
Sudhnoti	50.16%	73.8	50.0	85.4
Tor Ghar	43.64%	52.0	57.0	51.0

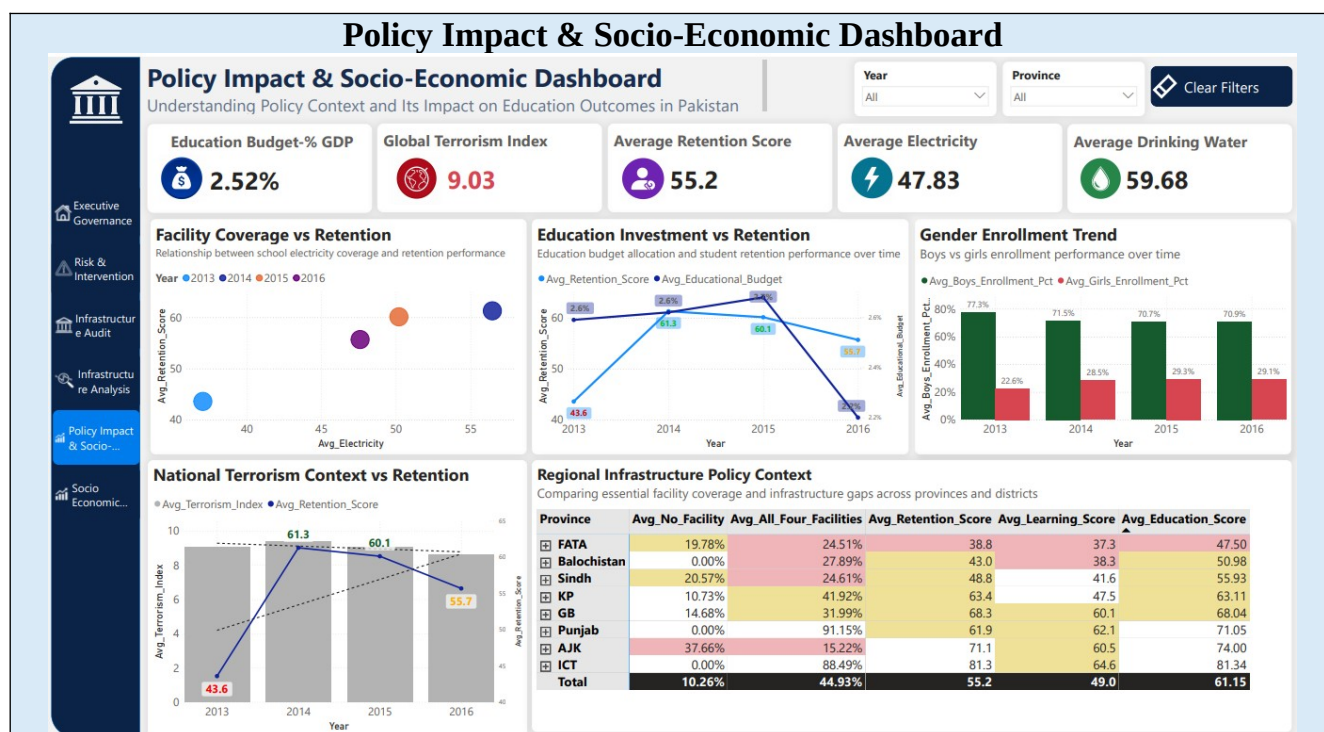
The results show that infrastructure deficit alone does not explain educational outcomes. Sudhnoti is a useful contrast because its No Facility value is high while retention is comparatively strong.

Recommended Government Action

Prioritize integrated interventions in Kohistan, Thatta and Tharparkar, combining essential facility improvement with teacher availability, classroom capacity, learning support and student-retention measures. Investigate stronger districts such as Sudhnoti for potentially transferable practices.

8. Dashboard 3 — Policy Impact & Socio-Economic

This dashboard places education performance in national policy and socio-economic context.



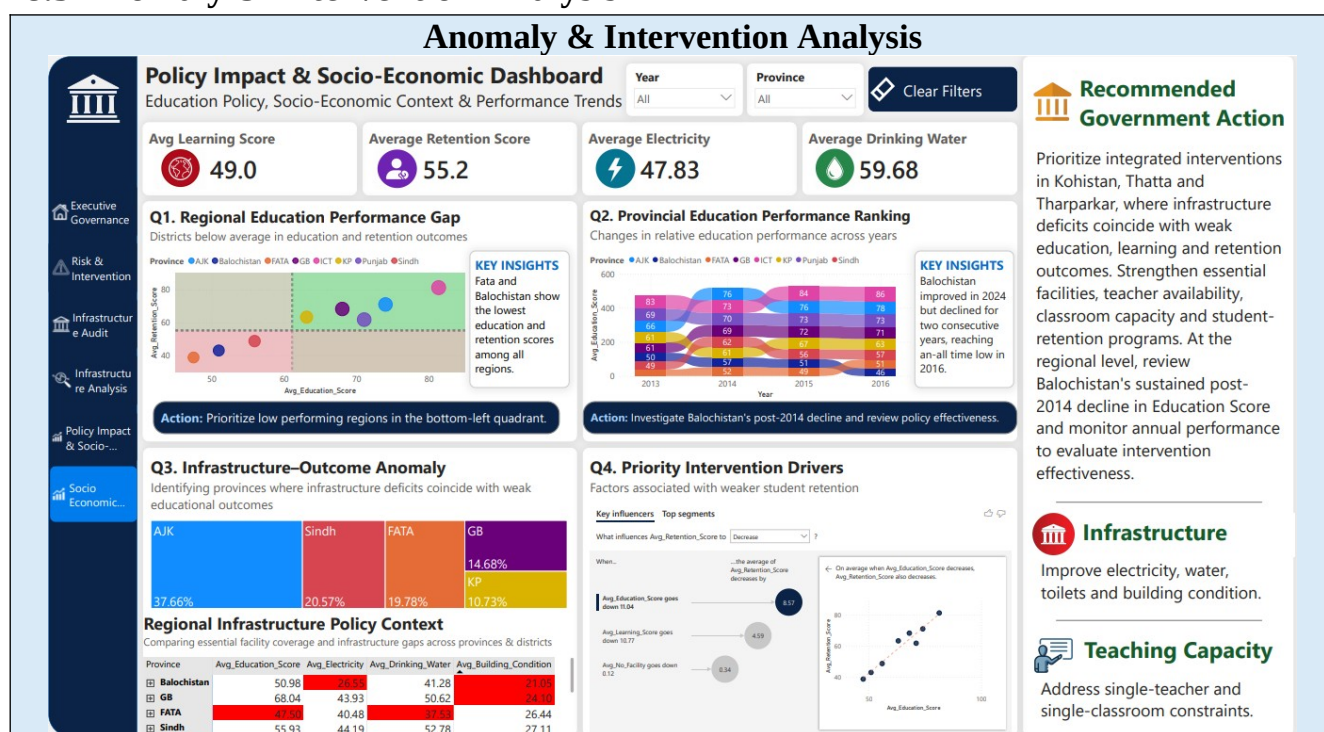
8.1 Questions Answered

- How does education performance change over time?
- How does national education investment change over time?
- How does retention change alongside national context?
- What is the relationship between facility coverage and retention?
- How does gender enrollment change?
- How does terrorism context change alongside retention?
- Which regional infrastructure conditions require investigation?

8.2 National-Level Context

Educational Budget Spend of GDP and Global Terrorism Index are treated as national-level context indicators. They respond to Year but are not interpreted as province-specific values.

8.3 Anomaly & Intervention Analysis



Q1 — Regional Education Performance Gap

Question: Which regions show comparatively weak education, learning and retention outcomes?

Region	Education	Learning	Retention
FATA	47.5	37.3	38.8
Balochistan	51.0	38.3	43.0
Sindh	55.9	41.6	48.8
KP	63.1	47.5	63.4
GB	68.0	60.1	68.3
Punjab	71.1	62.1	61.9

AJK	74.0	60.5	71.1
ICT	81.3	64.6	81.3

Recommended action: Prioritize further district-level investigation in weaker-performing regions, particularly FATA, Balochistan and Sindh, combining outcome monitoring with infrastructure and staffing analysis.

Q2 — Policy Performance Reversal

Question: Which province demonstrates a notable reversal in education performance over time?

Province	2013	2014	2015	2016	Pattern
AJK	66.2	75.7	76.0	78.1	Sustained improvement
Sindh	48.9	61.7	56.2	56.9	Improvement, decline, stabilization
Balochistan	49.8	57.2	50.8	46.1	Improvement followed by two years of decline

Key finding: Balochistan is the strongest policy-performance reversal candidate. Its score improved in 2014, then declined in 2015 and 2016, ending below its 2013 level.

Recommended action: Conduct a targeted policy and implementation review in Balochistan, examining district-level resource allocation, infrastructure, teacher capacity and implementation effectiveness. Monitor annual performance after interventions.

Q3 — Infrastructure–Outcome Anomaly

Question: Which districts combine infrastructure deficits with weak educational outcomes?

District	Province	No Facility	Education	Learning	Retention
Kohistan	KP	37.89%	40.1	42.1	30.0
Thatta	Sindh	44.42%	45.6	36.7	24.3
Tharparkar	Sindh	52.89%	51.1	34.8	33.1

Recommended action: Prioritize integrated intervention in Kohistan, Thatta and Tharparkar, combining essential facility improvements with teacher, classroom, learning and retention measures.

Q4 — Priority Intervention Drivers

Question: Which factors are associated with weaker student retention?

Influencer	Change	Associated Retention Change
Avg_Education_Score	Decreases by 11.04	Retention decreases by 8.57
Avg_Learning_Score	Decreases by 10.77	Retention decreases by 4.59
Avg_No_Facility	Decreases by 0.12	Retention decreases by 0.34

The strongest useful interpretation is that lower education and learning performance are strongly associated with weaker retention. These are statistical associations, not proof of causation.

Recommended action: Focus retention interventions on education and learning improvement, supported by infrastructure, staffing and district-level evidence.

9. Consolidated Recommended Government Actions

Priority Area	Evidence	Recommended Action
Gender Gap	Critical gaps; Kohistan showed 93.0%	Target girls' enrollment and participation measures and investigate access barriers.
Infrastructure	High No Facility in multiple districts	Improve electricity, water, toilets and building condition in priority districts.
Teacher Capacity	High single-teacher strain in risk areas	Review teacher deployment and prioritize high-strain districts.
Classroom Capacity	High single-classroom strain	Increase classroom capacity and reduce single-classroom constraints.
Learning	Lower learning is associated with lower retention	Implement targeted learning and foundational-skills support.
Retention	Weak retention in Kohistan, Thatta and Tharparkar	Develop district-specific retention interventions and monitor annually.
Balochistan Reversal	49.8 → 57.2 → 50.8 → 46.1	Conduct a focused policy and implementation review and monitor recovery.

10. Report Design Checklist Assessment

10.1 Overall Report Effectiveness

Checklist	Implementation	Status
Solves a business pain point	Centralizes education, infrastructure and policy information to identify intervention areas.	Achieved
Supports better decision making	Interactive filtering, risk analysis, drill-through, anomaly detection and actions.	Achieved
Impact can be measured	Potential measures include reduced analysis time, faster prioritization and improved monitoring.	Partially measurable / future evaluation
Adapted to intended audiences	Separate executive, risk, operational and policy views.	Achieved
Branding is consistent	Consistent titles, navigation, typography, KPI design and visual language.	Achieved
Professional design	Structured layouts, spacing, conditional formatting and varied advanced visuals.	Achieved
Visuals are readable	Visuals are selected according to analytical purpose with subtitles and context.	Achieved
UX is consistent	Common slicers, navigation, page structure and formatting.	Achieved
Metrics have context	Year/province filters, trends, reference lines and risk classifications.	Achieved

10.2 Information Architecture Effectiveness

Requirement	Implementation
Audience-aware views	Executive, risk, operational and policy-analysis pages match different personas.
Logical structure	Monitor → Identify → Investigate → Prioritize → Act.
Progressive detail	Users move from KPIs to province/city and district-level analysis.
Decision-oriented reporting	Each analysis page answers a defined question and leads toward action.

11. UX, Navigation and Interactivity

- Year and Province slicers
- Navigation buttons
- Province → City hierarchy
- Drill-through
- Conditional formatting
- KPI cards
- Advanced visuals
- Recommended Action panels

12. Limitations

- Data covers 2013–2016.
- National budget and terrorism indicators are contextual, not province-specific.
- Key Influencers identifies associations, not causation.
- Some district-level explanations require additional data.
- More recent government data would improve future versions.

13. Future Enhancements

- More recent data and automated refresh
- District-level education budgets
- Teacher deployment and attendance
- Student attendance and examination performance
- Socio-economic household indicators
- Forecasting and predictive analytics
- Automated anomaly alerts
- Role-based reporting

14. Conclusion

The project demonstrates how Power BI can transform complex education data into an interactive, stakeholder-driven decision-support solution. The report guides users from executive monitoring through anomaly detection, district investigation, prioritization, and recommended action.

The three dashboard areas—Executive Governance, District Infrastructure Audit and Policy Impact & Socio-Economic—provide complementary perspectives. Their analytical pages extend the solution with risk identification, infrastructure–outcome analysis, policy-performance reversal analysis and retention-driver analysis.

The final solution follows the principle: Data → Insight → Priority → Action. This aligns the report with the instructor's checklist by addressing a business pain point, supporting decision making, providing audience-specific views, maintaining consistent design and establishing a logical information architecture.